

Directions: It's been a while since covered a variety of topics from this course. This review worksheet covers *some* (not necessarily *all*) of the topics that we have learned. **DO NOT** take this review to be a copy of the final exam. Be sure to look over the Final Exam Study Guide and take heed of the other recommendations for preparing for the file. Remember that many of these problems are still *Challenge Problems* and therefore it is not expected that you might know how to answer each of these questions immediately.

1. Consider the region bounded by the curves $y = x^2 + 1$, $y = 1$ and $x = 3$.
 - a) Set up, but do not evaluate, an integral(s) that represents the volume of this solid created by rotating this region about the x -axis by integrating with respect to x .

 - b) Set up, but do not evaluate, an integral(s) that represents the volume of this solid created by rotating this region about the x -axis by integrating with respect to y .

 - c) Consider the solid formed with this region as its base and cross sections parallel to the x -axis are regular hexagons. Set up, but do not evaluate, an integral(s) that represents the volume of this solid.

2. A bullet is formed by revolving the area bounded by the curve $y = \ln(x)$ from $x = 1$ to $x = e$ about the x axis. It is then shot straight into a very thick wall and does not pierce through the other side. As such, it makes a cylindrical hole until it stops moving. Then the bullet is carefully extracted without affecting the hole at all, leaving an empty hole with a pointy end where the bullet once was. The length of the entire hole is $e+1$. Determine the volume of the hole created by the bullet.

3. Compute each of the following integrals.

a) $\int e^{\sqrt{x}} dx$ (*Hint: Start by substituting $x = y^2$*)

b) $\int \frac{dx}{(5 + 2x + x^2)^{3/2}}$

c) $\int \frac{\sin^3(\theta)}{\cos^2(\theta)} d\theta$

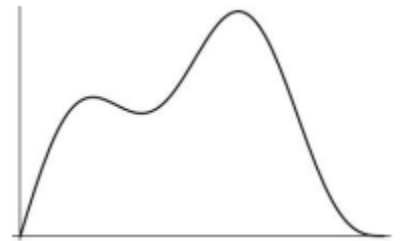
d) $\int \frac{x^3 + x - 1}{x^2 - 1} dx$

4. The volume of the solid formed by revolving the curve $y = \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{1+x}}$ bounded between $x = \frac{1}{8}$ and $x = \frac{9}{16}$ around the x -axis can be expressed in the form $A\pi \ln\left(\frac{B}{C}\right)$. If A and B contain no factors that are perfect squares, then find the value of $A \times B \times C$.

5. Evaluate the integral $\int_{-1}^1 \frac{1}{x} dx$.

6. Consider the curve $y = 2\sin(x) + \sin(2x)$, $0 \leq x \leq \pi$ shown at the right.

a) Set up, but do not evaluate, an integral which represents the length of the curve.



b) Set up, but do not evaluate, an integral which represents surface area obtained by rotating the curve about the x -axis.

c) Set up, but do not evaluate, an integral which represents surface area obtained by rotating the curve about the y -axis.

7. Recall that the Monotone Sequence Theorem test states that if a sequence a_n is (i) *increasing* and (ii) *bounded* then it must converge. Consider the sequence

$$a_1 = \sqrt{2}$$

$$a_2 = \sqrt{2 + \sqrt{2}}$$

$$a_3 = \sqrt{2 + \sqrt{2 + \sqrt{2}}}$$

...

$$a_{n+1} = \sqrt{2 + a_n}$$

- a) Show that if $a_n < 2$ then $a_{n+1} < 2$. Since $a_1 = \sqrt{2} < 2$ what does this implies about all values of a_n ?

- b) Show that a_n increasing.

- c) Conclude that a_n must converge. Can you see what it must converge to?

8. Determine the limit of the sequence $a_n = (-1)^n \arctan(n)$.

9. Determine if the following series converges or diverges $\sum_{n=2}^{\infty} (-1)^n n e^{-n}$

10. Determine if the each of the following series is absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

b) $\sum_{n=2}^{\infty} (-1)^n \frac{n^2}{n^3 - 1}$

c) $\pi^2 + \pi + 1 + \frac{1}{\pi} + \dots$

d) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{\ln(n+1)}$

11. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and that $\sum b_n$ converges.

- a) Show that if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ then $\sum_{n=1}^{\infty} a_n$ also converges. (*Hint:* Start by showing that the inequality $a_n \leq b_n$ is eventually true). (Note that this question is exploring an extension of the Limit Comparison Test.)

- b) Use part (a) to show that $\sum_{n=1}^{\infty} \frac{[\ln(n)]^2}{n^3}$ converges.

- c) Give an example of a pair of series $\sum a_n$ and $\sum b_n$ with positive terms where $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ diverges, but $\sum a_n$ converges.

12. Find the Taylor series (centered at the given point) for each of the following functions by differentiating the given functions and finding the pattern.

a) $f(x) = \sin(x)$ centered at $x_0 = \frac{\pi}{2}$

b) $f(x) = e^{3x}$ centered at $x_0 = 2$

13. Find the Radius and Interval of Convergence of the following power series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

b) $\sum_{n=0}^{\infty} \sqrt{n} x^n$

$$\text{c) } \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\text{d) } \sum_{n=0}^{\infty} \frac{(-1)^n (x-3)^n}{2n+1}$$

14. Give the Taylor polynomial of degree 2 centered at 4 for $f(x) = \frac{1}{\sqrt{x}}$. Use Taylor's Theorem to estimate the accuracy this approximation for x between 3.8 and 4.2.

15. Consider the curve $r = \cos(3\theta)$

a) Make a sketch of this curve and find the total area enclosed by the curve.

b) Set up an integral for the total length of the curve.

c) Determine the total area enclosed by this polar curve.

16. The coordinates of a dot P moving in the xy -plane at time t are given by:

$$x = 5e^{-t} \cos(t) \text{ \& } y = 5e^{-t} \sin(t)$$

Let L_a be the distance traveled by P for $0 \leq t \leq a$. If $\lim_{a \rightarrow \infty} L_a = z$, then determine the value of z .

17. Identify each of the following as either an ellipse, hyperbola, or parabola by putting each equation into standard form.

a) $x^2 + y^2 - 10x + 9 = 0$

b) $y^2 + 16x - 4y - 52 = 0$

c) $11x^2 - 25y^2 + 22x + 250y - 889 = 0$

